
3AL Test 1 2023 Solutions

Time: 90 minutes

Available marks: 70

Full marks: 65

Instructions

1. Answer all questions in the lined answer booklets.
2. You can answer the questions in any order but start each question on a new page.
3. Make sure each answer is clearly labelled.
4. Show all work and reasoning unless otherwise instructed.
5. This test consists of **4** questions. Make sure your question paper is complete.
6. If you have read the instructions, draw a star on the front of your answer booklet.
You will receive an extra mark for doing so.

Good luck! $\diamond_+^+$

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1. Let G_1 and G_2 be groups. Use the First Isomorphism Theorem to prove that there is a factor group of $G_1 \times G_2$ that is isomorphic to G_1 . Clearly show what the factor group is.

NOTE: This question is asking you to find a normal subgroup $H \triangleleft G_1 \times G_2$ such that $(G_1 \times G_2)/H \cong G_1$.

Consider the set $H = \{(1_{G_1}, g_2) : g_2 \in G_2\} \subseteq G_1 \times G_2$. We claim that $H \triangleleft G_1 \times G_2$ and $(G_1 \times G_2)/H \cong G_1$. We define a map $\varphi : G_1 \times G_2 \rightarrow G_1$ by $\varphi(g_1, g_2) = g_1$. For $(g_1, g_2), (g'_1, g'_2) \in G_1 \times G_2$, we have

$$\varphi((g_1, g_2)(g'_1, g'_2)) = \varphi(g_1 g'_1, g_2 g'_2) = g_1 g'_1 = \varphi(g_1, g_2) \varphi(g'_1, g'_2).$$

Thus φ is a homomorphism. Moreover, for any $g_1 \in G_1$, we have $(g_1, 1_{G_2}) \in G_1 \times G_2$ such that $\varphi(g_1, 1_{G_2}) = g_1$. Thus φ is an epimorphism. We have $(g_1, g_2) \in \ker \varphi$ if and only if $\varphi(g_1, g_2) = 1_{G_1}$. That is, $(g_1, g_2) \in \ker \varphi$ if and only if $g_1 = 1_{G_1}$. Thus $\ker \varphi = H$. Then by the First Isomorphism Theorem, $(G_1 \times G_2)/H \cong G_1$ (the normality of H also follows from the theorem).

[15 marks]

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2. (a) State the Second Isomorphism Theorem.
- (b) Let G be a group of order $p^n q$, where p and q are distinct primes and $n \geq 1$. Let K be a normal subgroup of G of order p^n . Suppose H is a subgroup of G .
- (i) Show that $|HK/K|$ divides q .
- (ii) Show that if $|H| = p^n$, then $|HK/K| = 1$ and $|H \cap K| = p^n$.
- (iii) Explain why K is the only subgroup of G with order p^n .

(a) **Second Isomorphism Theorem.** *Let H and K be subgroups of G with $K \triangleleft G$. Then $H/(H \cap K) \cong (HK)/K$.*

(b) (i) By the Correspondence Theorem, $HK/K \leq G/K$. By Lagrange's Theorem, $|G/K| = p^n q / p^n = q$. Then, by Lagrange's Theorem again, $|HK/K|$ must divide $|G/K| = q$.

(ii) By the Second Isomorphism Theorem, $H/(H \cap K) \cong HK/K$. Thus the orders of these groups must be equal. Since $H \cap K$ is a subgroup of H , its order must divide that of H . That is, $|H \cap K|$ is a power of p . Then $|H|/|H \cap K| = p^k$ for some $0 \leq k \leq n$. Then $|HK/K| = p^k$. But $|HK/K|$ divides q , which is not divisible by p . So, we must have $k = 0$. That is, $|HK/K| = 1$. Moreover, $|H \cap K| = |H|/p^k = |H|/1 = p^n$.

(iii) If there was another subgroup H of G with order p^n , then we would have $|H \cap K| = p^n$. But $H \cap K$ is a subgroup of H , which also has order p^n . Then $H \cap K = H$. Then $H \subseteq K$ but $|H| = |K|$ and so $H = K$. Thus, K is the only subgroup of G with order p^n .

[5 + 5 + 10 + 5 = 25 marks]

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3. Let G be a group with normal subgroups H and K such that $H \subseteq K$. Prove or disprove: If G/H is cyclic, then so are G/K and K/H .

Since G/H is cyclic, so are its subgroups. Thus, K/H is cyclic. Moreover, since G/H is cyclic, so are its factor groups. By the Third Isomorphism Theorem, $G/K \cong (G/H)/(K/H)$. Thus $(G/H)/(K/H)$ is cyclic and then so is G/K , since it is isomorphic to the aforementioned group.

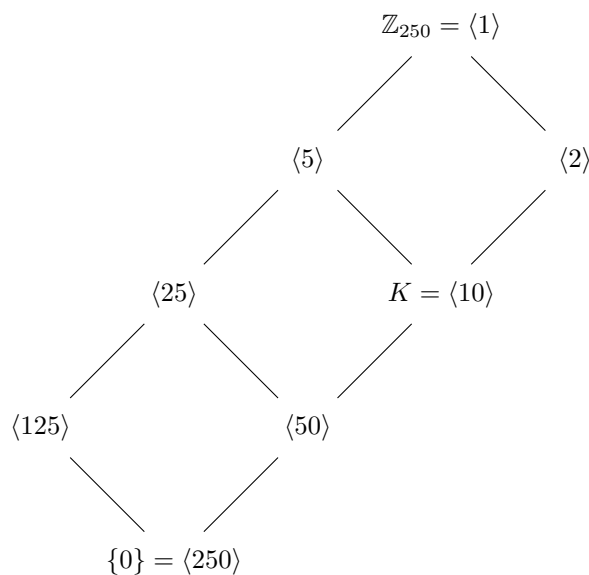
[10 marks]

4. Consider the group $G = \mathbb{Z}_{250}$ with subgroup $K = \langle [10] \rangle$. Note that $250 = 2 \cdot 5^3$.

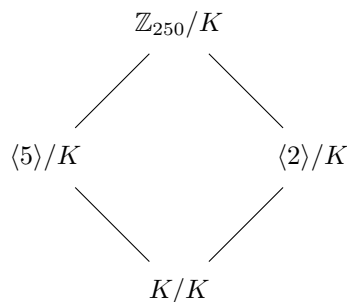
- (a) Determine all subgroups of $\mathbb{Z}_{250}$ and draw the lattice diagram.
- (b) Use the Correspondence Theorem to draw the lattice diagram for $\mathbb{Z}_{250}/K$.
- (c) Let $H = \langle [50] \rangle$. Determine the number of subgroups of $(\mathbb{Z}_{250}/H)/(K/H)$.

NOTE: You can use the fact that isomorphic groups have the same number of subgroups (no need to prove this).

- (a) By the Fundamental Theorem of Finite Cyclic Groups, the subgroups of $\mathbb{Z}_{250}$ correspond to the divisors of 250. Namely, 1, 2, 5, 10, 25, 50, 125 and 250. The subgroups of $\mathbb{Z}_{250}$ are the unique subgroups groups of these orders. Namely, $\{0\} = \langle 250 \rangle$, $\langle 125 \rangle$, $\langle 50 \rangle$, $\langle 25 \rangle$, $\langle 10 \rangle$, $\langle 5 \rangle$, $\langle 2 \rangle$, $\mathbb{Z}_{250} = \langle 1 \rangle$. The lattice diagram is:



- (b)



(c) By the Third Isomorphism Theorem, $(\mathbb{Z}_{250}/H)/(K/H) \cong \mathbb{Z}_{250}/K$. In the above lattice diagram, we see that $\mathbb{Z}_{250}/K$ has 4 subgroups. Thus, $(\mathbb{Z}_{250}/H)/(K/H)$ also has 4 subgroups.

[10 + 5 + 5 = 20 marks]